

Senior Engineer for 10+ years building award-winning systems across fintech, healthcare, and national security.

Founder & AI Engineer ■ Spirit & Hammer

October 2025 - Present

Technologies: Next.js, React, Angular, TypeScript, Python, Python-Django, FastAPI, Playwright, Claude, Vercel, Cloudflare, AI/LLMs

- Architected full-stack features and end-to-end AI agent harnesses, MCP server integrations, and agent skills for SOC startup Intruex and its Python-Django application, contributing to its first government contract.
- Shipped Sun Signal, a circadian-health platform delivering real-time daily timing guidance for thousands of visitors across US ZIP codes, using Next.js, AI chat, dynamic Vercel AI UI components, and a 50K-LOC Python FastAPI astronomy data engine.
- Architected Next.js and Python-based AI video pipelines and a YouTube player system that protected user agency and attention while reducing video-production time by 50%.
- Delivered a Next.js personal-brand platform with blogs, newsletters, lead capture, and attribution inside a sandboxed natural-language builder, enabling a nontechnical client to develop features with AI.
- Independently prototyped Green Lights, a travel utility built with Next.js, Python, Google Cloud APIs, HMAC magic-link authentication, and Cloudflare.
- Engineered agentic-first, offline-first applications in Next.js, React, and Angular v20 on Vercel and Cloudflare, with encryption, distributed databases, offline functionality, email, newsletters, and custom HMAC magic-link authentication.

Principal Software Engineer ■ Capital One

Jan 2021 to Dec 2021, Aug 2024 to Oct 2025

Technologies: Java, JavaScript, Python, TypeScript, SolidJS, React, Angular (v1-v14), Spring Boot, AWS, Docker, Git, REST, AI, Copilot

- Generated millions in revenue by delivering Auto Navigator Trade-In in SolidJS within two months on an understaffed team, leading cross-functional alignment and owning a scalable document-upload feature.
- Built full-stack Trade-In, Auto Loan Rates, Search, and Capital One Banking features serving millions of users with React, SolidJS, Angular v1-v14, Python, and Spring Boot microservices on AWS.
- Built Python backend APIs and AWS Lambda serverless services supporting Capital One Bank and SEO-facing Auto Loan Rates and Search pages.
- Used DynamoDB for scalable NoSQL storage, Fargate and EC2 for compute, S3 for application bundles and SEO assets, and custom internal AWS tooling for resource management.
- Cut code-review cycles by 40% and eliminated 100,000 API failures by leading cybersecurity remediation and automated security testing as Duty Champion.
- Enabled zero-defect deployments for more than 3 million daily users by modernizing legacy Angular v1/AngularJS code and strengthening automated testing and release processes.
- Won a company-wide hackathon for UX design and frontend implementation.

Senior Software Engineer ■ Publicis Sapient

April 2022 to May 2023

Technologies: Angular (v12 - v14), TypeScript, Java, Spring Boot, AWS, ArcGIS, Docker, REST APIs, JUnit

- Built interactive healthcare-facility mapping with ArcGIS and Angular v12-v14, helping officials map providers across regions and increasing healthcare access by 20% in underserved US states.
- Standardized code reviews and led mentorship programs, reducing defect rates by 33% and establishing quality benchmarks adopted company-wide.
- Designed Angular Material component libraries that standardized cross-team design patterns and accelerated feature delivery.

Senior Software Engineer ■ Maxar Technologies

Dec 2018 to Apr 2022

Technologies: Angular (v4 - v12), TypeScript, Java, Spring Boot, AWS, Kubernetes, REST APIs, JUnit

- Supported national decision-makers by engineering distributed Angular v4-v12 applications for geospatial mapping and probability simulation.
- Enabled rapid analysis of high-volume data by building real-time visualizations with parallel processing in modern JavaScript frameworks.
- Partnered with technical and business stakeholders to validate requirements, deliver technical demonstrations, and drive adoption of complex engineering solutions.

Software Engineer ■ MITRE Corporation

Dec 2016 to Dec 2018

Technologies: Angular, Material Design, JavaScript, TypeScript, Python, Python-Django, Java, OAuth2, JWT, PostgreSQL, Git

- Built a Python-Django DOJ cyber-incident analysis platform, a million-dollar DoD cost-benefit analysis, and an Angular v2 MEAN-stack web IDE supporting the multiyear IRS legacy tax-system modernization effort.
- Implemented secure JavaScript, Typescript and Python-based OAuth2/JWT authentication and automated testing frameworks to strengthen enterprise security, compliance and release confidence.
- Researched automated testing, software-quality metrics, blockchain, and commercial and government cloud providers supporting sponsor decision-making.

Software Engineer ■ Americas Cooperatives

Jan 2015 to Dec 2016

Technologies: *Python, Flask, JavaScript, jQuery, HTML/CSS, Apache, GridLAB-D, PVWatts, AWS, Travis CI, Git*

- Developed Python-Flask open-source power-system cost-benefit models for the Open Modeling Framework, informing million-dollar grid investments with projected energy savings over 50% and ROI supporting cooperative enrollment and clean-energy adoption.
- Designed model architecture, visualizations, and ETL workflows in Python and Julia while managing code and onboarding graduate researchers.
- Independently prototyped a Python-based gamified spear-phishing awareness simulator.

Languages: Python, Java, Typescript, JavaScript, Node.js, HTML/CSS**AI & ML:** Agentic AI, RAG Pipelines, ML & LLM, Claude & Claude Code, Claude Certified Architect (August 2026), OpenRouter, AI SDK**Frontend:** React, Angular v1/AngularJS to v20, RxJS & NgRx, geospatial mapping (ArcGIS, Cesium.js, Leaflet.js, Google Maps API)

Firebase/Firestore, Nx monorepo

Backend: Python, Flask, Django, FastAPI, Spring, Node.js REST, Microservices, MongoDB, NoSQL, distributed systems, parallel computing**Cloud, DevOps & Testing:** AWS Certified Solutions Architect, Docker, Kubernetes, CI/CD, Git, JUnit/Jest/Selenium, TDD/BDD**Certifications:** Claude Certified Architect (August 2026), John Hopkins Agentic AI, AWS Certified Solutions Architect (July 2025)

Bachelor of Science, Electrical Engineering, George Mason University